RAttAGS: X-ray Attenuation-Aligned 3D Gaussian Splatting for Sparse Tomographic View Synthesis

Abstract

3D Gaussian Splatting has been extended to X-ray novel view synthesis, yet existing methods retain the adaptive density control from natural-image 3DGS without modification. Under sparse CT acquisition, these rules cause Gaussians to accumulate near high-contrast anatomical boundaries while interior tissues that determine X-ray attenuation remain under-represented. We present ***RAttAGS****, an X-ray Attenuation-Aligned 3D Gaussian Splatting framework that corrects this structural imbalance by redesigning the adaptive density control pipeline. RAttAGS introduces three components: a support-profile seeding strategy that initializes Gaussians from the coarse attenuation support estimated by FDK reconstruction; a geometry-aware pruning step that removes Gaussians concentrated near anatomical boundaries; and an adaptive density modulation step that applies position-dependent density refinement throughout training. Experiments on five organs under a sparse-view CT protocol demonstrate that RAttAGS achieves state-of-the-art novel view synthesis quality against six representative baselines from both visible-light sparse-view synthesis and X-ray CT methods.*

1. Introduction

Computed tomography (CT) acquires multi-angle X-ray attenuation measurements to reconstruct an object's internal structure, and is indispensable in clinical diagnosis, image-guided intervention, and industrial inspection [FDK84; AK84; WYD08]. Reducing the number of acquired views directly lowers radiation dose, shortens scan time, and broadens applicability to patients who cannot sustain long acquisitions [LMC*23; WHL*24]. In this work, we study *Sparse Tomographic View Synthesis*: from only a handful of X-ray projections, the model must predict the line-integrated attenuation at unseen acquisition angles, so that each synthesized projection remains consistent with the Beer–Lambert imaging geometry. A reliable solution supplies auxiliary views for anatomical localization and intraoperative planning where conventional analytical reconstruction breaks down [WN19].

Visible-light imaging captures surfaces through reflection and scattering, whereas X-ray imaging reveals internal structures through penetration and attenuation—two fundamentally different physical processes that dictate where representation capacity should reside. In visible-light 3DGS, a pixel is formed by front-to-back α-compositing [KKLD23; Max95]: transmittance decays multiplicatively, so for opaque scenes the integrand is dominated by Gaussians near the first-hit surface—a transmittance-induced concentration consistent with surface-aligned 3DGS variants [GL24; HYC*24]. In X-ray imaging, the detector reading follows the Beer–Lambert law [KS01]: after taking the logarithm, the recorded quantity becomes a linear, order-independent line integral of the attenuation coefficient, with no transmittance decay and no occlusion. Every Gaussian on the path therefore contributes on equal footing, and capacity must be distributed *along the entire ray path*. Existing X-ray Gaussian methods have correctly replaced α-compositing with a radiative line-integral renderer [CLW*24; ZLC*24; WTW*25; WLG*25], and sparse-view 3DGS pipelines further compensate for missing observations through depth or co-regularization priors [ZFJW24; ZLY*24; LZB*24]. Yet all of them inherit an adaptive density control strategy originally designed for surface-centric α-compositing [KKLD23; RNSZ24].

Crucially, the issue is not gradient-driven densification itself—splitting or cloning Gaussians where the projection-error gradient is large is sound, and our method retains this mechanism. The issue is that under a line-integral renderer, the residual gradient is still concentrated at high-attenuation-contrast interfaces such as bone–soft-tissue boundaries. A naïve port therefore keeps spawning Gaussians at anatomical boundaries even though the rendering equation asks for capacity *along* the path. Under sparse views, this bias manifests as a structural *capacity misallocation*: Gaussians accumulate at boundaries while interior path segments that determine the attenuation integral remain under-represented, as illustrated in Fig. 1. Fixing this requires *augmenting*, not discarding, gradient-driven densification with attenuation-aligned decisions about where to seed, what to prune, and how to refine.

We propose **RAttAGS**, an X-ray Attenuation-Aligned 3D Gaussian Splatting framework for Sparse Tomographic View Synthesis. **RAttAGS** keeps the gradient-driven backbone of standard 3DGS and adds three attenuation-aligned stages that act on *where* Gaussians are seeded, retained, and refined. Support-Profile Seeding (SPS) initializes Gaussians from a coarse FDK attenuation support [FDK84] so that they populate physically meaningful X-ray paths; Geometry-aware Pruning (GAP) recycles Gaussians that accumulate near high-contrast anatomical interfaces, draining boundary-driven redundancy at its source; and Adaptive Density Modulation (ADM) applies position-dependent online refinement so that retained Gaussians carry interior attenuation rather than uniformly inflated density.

Our main contributions are summarized as follows:

- We identify *capacity misallocation* as the core failure mode when sparse-view X-ray Gaussian methods inherit visible-light adaptive density control under a line-integral renderer.
- We propose **RAttAGS**, an X-ray Attenuation-Aligned 3D Gaussian Splatting framework that augments gradient-driven densification with three attenuation-aligned stages—SPS for path-anchored initialization, GAP for boundary-redundancy recovery, and ADM for position-dependent density refinement—without modifying the X-ray renderer.
- Experiments on multi-organ sparse-view CT

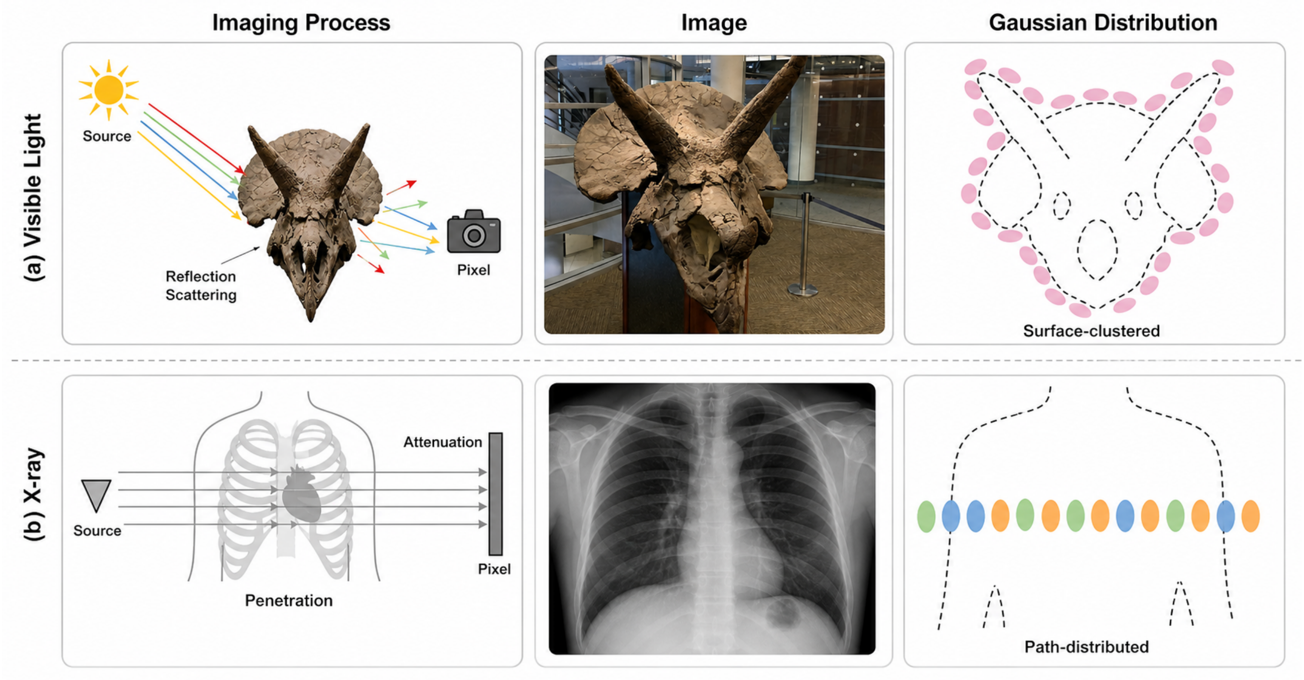

Figure 1: ***Visible-light vs. X-ray Gaussian rendering and the capacity gap they impose.*** *(a) Visible-light α-compositing concentrates capacity on visible surfaces, where front Gaussians occlude back ones. (b) X-ray line integrals weight every Gaussian along the ray equally, so naïve gradient-driven densification still clusters at attenuation boundaries;* ***RAttAGS*** *realigns the density control strategy through SPS, GAP, and ADM to anchor capacity along the path.*

show that **RAttAGS** delivers consistent gains over state-of-the-art baselines; ablations confirm that SPS, GAP, and ADM each contribute complementary capacity reallocation.

2. Related Work

2.1. Gaussian Splatting

3D Gaussian Splatting (3DGS) represents a scene with a set of anisotropic Gaussians and renders novel views in real time through differentiable tile-based rasterization [KKLD23]. Building on this explicit representation, subsequent work has expanded along several directions: surface-aligned variants that recover geometrically accurate meshes from Gaussians [HYC*24; GL24]; dynamic and deformable extensions for time-varying scenes [WRW*24; YGZ*24]; dense visual SLAM systems built on top of the Gaussian representation [YQX*24]; generative pipelines that lift 3DGS into 3D content creation [TRZ*24]; and compactness/efficiency redesigns that reduce the Gaussian count or stabilize rendering across scales [NMKW24; LYX*24; YCH*24]. Across all these directions, the underlying adaptive density control remains the gradient-driven clone-and-split strategy together with spherical-harmonics view-dependent color, both of which are tailored to dense-view, surface-reflectance scenes. When either assumption is violated—inputs become sparse, or the imaging model switches from surface radiance to line-integral attenuation—this strategy starts to misallocate Gaussian capacity, and the next two subsections examine these two regimes in turn.

2.2. Sparse View Synthesis

Neural Radiance Fields (NeRF) recover a continuous volumetric field through coordinate MLPs and volume rendering [MST*21], and degrade noticeably when only a handful of input views are available. NeRF-side sparse-view methods approach this limitation from complementary angles, each adding a different external prior that has been empirically validated: depth supervision from sparse structure-from-motion points [DLZR22; WCLL23], semantic consistency from pretrained vision encoders [JTA21], and geometric/appearance regularization with frequency-domain annealing [NBM*22; YPW23]. Sparse-view 3DGS work explores a similarly diverse set of perspectives on the explicit Gaussian set: FSGS attributes degradation to insufficient coverage and augments densification with proximity-guided unpooling and depth priors [ZFJW24]; CoR-GS treats it as unstable optimization and adds dual-field co-regularization with co-pruning [ZLY*24]; DNGaussian injects geometry through depth normalization [LZB*24]; and DropGaussian counters overfitting via stochastic Gaussian dropout [PRK25]. A complementary feed-forward line further moves geometry completion into large-scale pretraining [CLTS24; CXZ*24; XPW*25; LFY*25]. These different angles all yield empirical gains, which suggests that sparse-view degradation admits multiple compatible explanations rather than a single root cause. The next subsection turns to how that assumption itself is challenged under X-ray line integration.

2.3. Tomographic Novel View Synthesis

Sparse-view tomographic novel view synthesis aims to recover unseen X-ray projections, and the underlying density volume, from a small number of input projections [WYD08; WN19], in contrast to analytical (FDK) and iterative (SART) reconstructions that either suffer severe streak artifacts under sparse views or require many iterations [FDK84; AK84]. Learning-based methods fall into two families with complementary entry points. Implicit methods adapt neural attenuation fields to X-ray imaging—through self-supervised sinogram synthesis, hash-encoded acceleration, transformer-based structural modeling, or population-level encoder–decoder inference [ZIL*21; GWS*24; CWY*24; LFL*25]. Explicit methods redesign Gaussians around X-ray attenuation, covering radiative kernels with X-ray rasterization, integration-bias correction with differentiable voxelization, and material-aware ellipsoids with explicit segment-length physics, alongside further extensions toward discretized voxels, artifact-suppression graphs, 4D dynamics, and slice-wise reconstruction [CLW*24; ZLC*24; WTW*25; WLG*25; YMS*25; YCZ*25; Kua*25] (the last four target imaging regimes outside our sparse-view novel-view protocol and are omitted from the main comparison). Across both families, improvements have concentrated on the imaging equation or the kernel itself rather than on the density control strategy that decides where Gaussians accumulate, leaving sparse-view CT still bottlenecked by capacity misallocation inherited from visible-light densification. **RAttAGS** addresses this density control strategy directly, with SPS, GAP, and ADM aligning initialization, pruning, and refinement to the X-ray attenuation path.

3. Method

3.1. Preliminary

3.1.1. X-ray Attenuation Imaging

An X-ray detector pixel records the residual energy of a single ray after traversing the object, governed by the Beer–Lambert law. For a ray $\mathbf{r}$ with source intensity I_0 and attenuation field $\mu(\mathbf{x})$, the standard practice is to work in the logarithmic domain [KS01], so that each pixel becomes a line integral of attenuation along the ray:

$$I(\mathbf{r}) = -\log \frac{I'(\mathbf{r})}{I_0} = \int_{\mathbf{r}} \mu(\mathbf{x}) \, \mathrm{d}l. \tag{1}$$

Unlike α-compositing in visible-light rendering, this integral is linear and order-independent: every point along $\mathbf{r}$ contributes additively, with no occlusion between front and back materials.

3.1.2. Radiative Gaussian Backbone

We build upon the R2-Gaussian framework [ZLC*24], which represents the scanned object with a set of radiative Gaussians $\mathcal{G} = \{G_i\}_{i=1}^{N}$. Each Gaussian G_i is parameterized by a center $\mathbf{p}_i$, a covariance Σ_i, and a non-negative density ρ_i, defining a local density field

$$G_i(\mathbf{x}) = \rho_i \exp\Big(-\tfrac{1}{2}(\mathbf{x}-\mathbf{p}_i)^\top \Sigma_i^{-1} (\mathbf{x}-\mathbf{p}_i) \Big), \tag{2}$$

and the global attenuation field is the sum $\mu(\mathbf{x}) = \sum_{i=1}^{N} G_i(\mathbf{x})$. Substituting this representation into Eq. (1) admits a differentiable X-ray rasterizer $\mathcal{R}$, so that for any acquisition geometry π_v a projection can be rendered as

$$\hat{I}_v = \mathcal{R}(\mathcal{G}; \pi_v). \tag{3}$$

3.1.3. Problem Statement

Given a sparse-view observation set $\mathcal{P} = \{(I_v, \pi_v)\}_{v=1}^{V}$ and a coarse FDK reconstruction V_{FDK}, our goal is to learn $\mathcal{G}$ in a per-case optimization setting so that target projections at unseen angles can be synthesized from only a few observed projections. The radiative backbone has already aligned the renderer with X-ray physics; what remains misaligned in this regime is the density control strategy that shapes $\mathcal{G}$ during training, where adaptive density control continues to concentrate Gaussians at high-gradient anatomical boundaries. **RAttAGS** realigns this density control strategy through three modules introduced below, while leaving the rasterizer and the differentiable voxelizer of R2-Gaussian [ZLC*24] unchanged.

3.2. Overall Framework

As illustrated in Figure 2, **RAttAGS** instantiates the attenuation-aligned density control pipeline as three modules acting at three distinct stages of optimization: SPS replaces standard initialization by seeding Gaussians along the FDK attenuation support; GAP acts inside the densification loop and reclaims locally redundant Gaussians produced by gradient-driven growth; ADM acts on retained Gaussians and refines their density values using position-dependent spatial context. At inference, the rasterizer remains identical to the radiative backbone, so the added cost is concentrated in training.

3.3. Support-Profile Seeding (SPS)

Initialization strongly conditions the optimization trajectory of radiative Gaussians, yet standard 3DGS initializers transfer poorly to X-ray tomography: uniform random sampling ignores the strong contrast between air, soft tissue, and bone, while SfM point-cloud initialization is unstable under gray-value overlap and weak local texture in X-ray projections [CLW*24]. X-Field [WTW*25] addresses this by progressively refining a coarse CT estimate

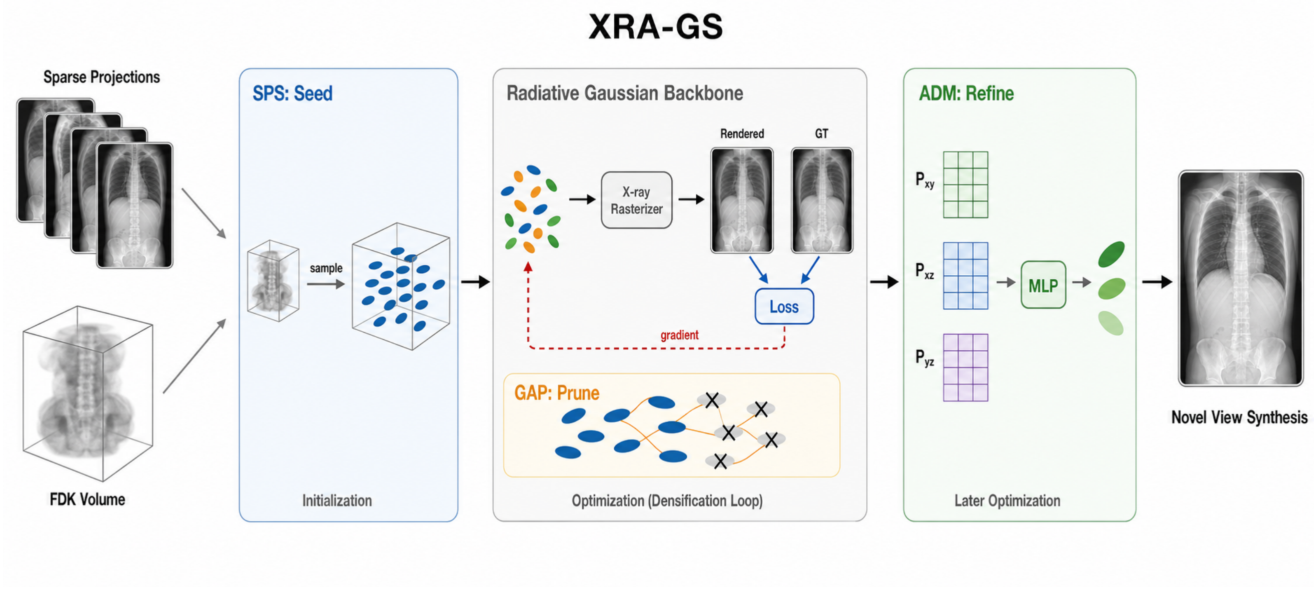

Figure 2: ***Overall framework of RAttAGS.*** *Sparse X-ray projections together with a coarse FDK reconstruction enter a radiative Gaussian backbone whose adaptive density control is reorganized by SPS (seed), GAP (prune), and ADM (refine). The three modules act at initialization, inside the densification loop, and during later optimization, respectively, leaving inference unchanged.*

through a CGLS–SART–TV pipeline before initializing ellipsoids at the reconstructed voxel centers with random covariance and attenuation attributes. While effective, this multi-stage iterative reconstruction adds considerable overhead. We observe that a single-pass FDK volume V_{FDK}, despite streak artifacts in fine detail, already provides reliable foreground support and low-frequency attenuation layout [WCS*22; LMC*23; LFZ*25; ZLC*24], which suffices as a sampling prior for the subsequent density control stages (GAP, ADM) to refine.

Concretely, over the voxel grid Ω underlying V_{FDK} we draw N seed positions $\{\mathbf{x}_i\}_{i=1}^{N}$ from a density-weighted mixture distribution

$$q(\mathbf{x}) = \alpha\,\frac{1}{|\Omega|} + (1-\alpha)\,\frac{V_{\mathrm{FDK}}(\mathbf{x})^{\gamma}}{Z}, \qquad Z = \sum_{\mathbf{x}'\in\Omega} V_{\mathrm{FDK}}(\mathbf{x}')^{\gamma}, \tag{4}$$

that interpolates between a uniform prior (weight α) and a power-shaped attenuation profile (exponent γ).

Each sampled position $\mathbf{x}_i$ then defines a Gaussian G_i with attributes initialized as:

$$\boldsymbol{\mu}_i = \mathbf{x}_i, \qquad \sigma_i = V_{\mathrm{FDK}}(\mathbf{x}_i), \qquad s_i = \frac{1}{K}\sum_{j\in\mathcal{N}_K(i)} \|\mathbf{x}_i - \mathbf{x}_j\|_2, \tag{5}$$

where $\boldsymbol{\mu}_i$ is the center position, σ_i the initial attenuation coefficient queried directly from the FDK volume, and s_i the isotropic scale set by the mean distance to the K-nearest seed neighbors $\mathcal{N}_K(i)$. The mixture form is critical: density weighting alone would collapse capacity onto the brightest bony structures, while the uniform term retains global coverage so that under-attenuating but anatomically relevant soft-tissue regions remain populated.

3.4. Geometry-aware Pruning (GAP)

Inside the radiative backbone, projection-error gradients are systematically larger near bone–soft-tissue boundaries, so standard gradient-driven densification

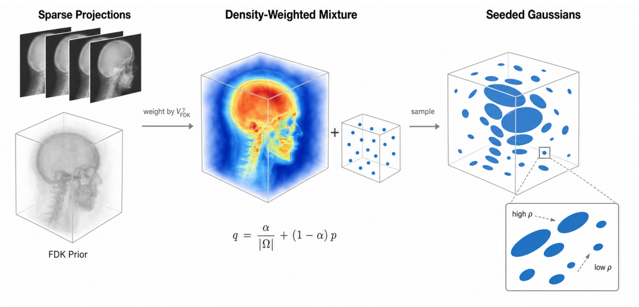

Figure 3: ***Support-Profile Seeding (SPS).*** *From the coarse FDK volume, SPS builds a density-weighted mixture distribution that interpolates between a uniform prior and the attenuation profile. Sampling from this mixture seeds Gaussians along the anatomical support before any optimization step.*

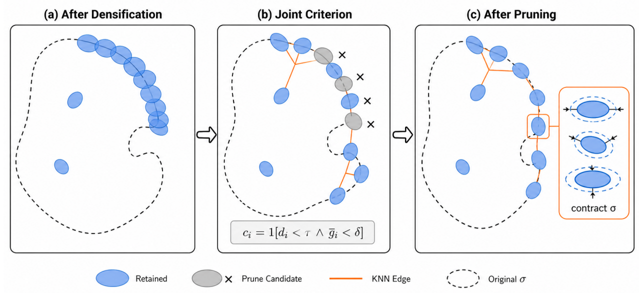

Figure 4: ***Geometry-aware Pruning (GAP).*** *After gradient-driven densification, Gaussians cluster around high-contrast boundaries. GAP flags a Gaussian as a redundancy candidate only when it is both locally crowded ($d_i < \tau$) and recently inactive ($\bar{g}_i < \delta$), so that capacity is reclaimed from saturated boundaries without removing Gaussians that are still learning structure.*

keeps duplicating Gaussians around the same high-contrast surfaces. Large gradients in this regime do not always imply missing coverage: they can also reflect that an already-saturated boundary is being represented redundantly by neighboring Gaussians. Rather than adding more Gaussians in sparse regions in the spirit of FSGS [ZFJW24], GAP acts in the opposite direction and reclaims locally crowded Gaussians during densification.

Following the proximity-graph construction of FSGS [ZFJW24], we connect each Gaussian G_i to its K nearest neighbors $\mathcal{N}_K(i)$ in position space and compute a proximity score as the mean neighbor distance:

$$d_i = \frac{1}{K} \sum_{j \in \mathcal{N}_K(i)} \|\mathbf{p}_i - \mathbf{p}_j\|_2. \tag{6}$$

A small d_i indicates that G_i sits in a locally crowded region. We combine the proximity score with $\bar{g}_i$, the accumulated positional-gradient magnitude of G_i over the current densification cycle, and flag a Gaussian as a redundancy candidate only when it is simultaneously locally crowded and recently inactive:

$$c_i = \mathbf{1}[d_i < \tau \ \wedge \ \bar{g}_i < \delta] \, , \tag{7}$$

where $c_i = 1$ marks a prune candidate. GAP removes a bounded fraction of the flagged Gaussians per cycle and slightly contracts the covariance of their retained neighbors to avoid creating local support holes. The dual criterion distinguishes GAP from purely density-based pruning: gradient activity prevents deleting Gaussians that are still learning a boundary, while proximity prevents over-densified regions from continuously monopolizing capacity.

3.5. Adaptive Density Modulation (ADM)

Even after path-anchored initialization and redundancy control, density values across anatomy remain heterogeneous: bone exhibits higher attenuation, air-filled regions should stay near zero, and soft tissue requires smoother transitions. A per-Gaussian gradient step updates each density value independently, making it difficult to capture the spatial neighborhood context needed to coordinate density allocation across these distinct tissue regimes. A straightforward remedy would be to employ a coordinate MLP to directly parameterize position-dependent density corrections, but such a network is prone to overfitting under sparse views where only a handful of projections constrain the entire volume. A dense 3D feature grid avoids this issue yet incurs memory that scales cubically with resolution. Inspired by the factorized plane representation of K-Planes [FYT*23], ADM decomposes the 3D volume into three orthogonal 2D feature planes, reducing memory to quadratic growth while preserving effective spatial neighborhood encoding. From these plane features, ADM decodes a position-dependent density correction for each Gaussian without modifying its center or covariance.

3.5.1. Spatial Feature Encoding

Given a Gaussian center $\mathbf{x}$, ADM projects it onto three orthogonal planes $\{P_{uv}\}_{uv\in\{xy,xz,yz\}}$, extracts per-plane features via bilinear interpolation at the corresponding 2D projections $\Pi_{uv}(\mathbf{x})$, and concatenates them into a joint spatial descriptor:

$$f_{uv}(\mathbf{x}) = \mathcal{B}\big(P_{uv},\, \Pi_{uv}(\mathbf{x})\big), \quad uv \in \{xy, xz, yz\}, \quad (8)$$

$$F(\mathbf{x}) = \big[f_{xy}(\mathbf{x});\ f_{xz}(\mathbf{x});\ f_{yz}(\mathbf{x})\big], \quad (9)$$

where $\mathcal{B}(\cdot)$ denotes bilinear sampling. The three-plane factorization implicitly encodes 3D spatial context: each sampled feature captures a 2D slice of the neighborhood, and their concatenation provides cross-plane spatial constraints that a single plane cannot express alone.

3.5.2. Density Modulation

The concatenated feature $F(\mathbf{x})$ is fed into a lightweight dual-head MLP h that outputs a density offset $\Delta\sigma(\mathbf{x}) \in [-1, 1]$ and a confidence gate $g(\mathbf{x}) \in [0, 1]$. To prevent ADM from degenerating into a global density scaling factor, we subtract the batch mean $\overline{\Delta\sigma}$ before applying a multiplicative modulation to the base density:

$$\rho_{\text{final}}(\mathbf{x}) = \rho_{\text{base}}(\mathbf{x}) \cdot \Big(1 + g(\mathbf{x}) \cdot \big(\Delta\sigma(\mathbf{x}) - \overline{\Delta\sigma}\big)\Big). \quad (10)$$

The plane features and the MLP parameters are trained end-to-end under the rendering loss alone, with no extra density supervision. ADM is activated only after the backbone has converged to a stable geometric estimate, so that density modulation refines an already reasonable layout rather than interfering with early-stage optimization. Batch centering ensures that $\Delta\sigma$ acts as a relative correction: ADM amplifies density where the local attenuation evidence warrants finer detail and suppresses modulation where evidence is insufficient.

3.6. Training Loss

RAttAGS is supervised end-to-end on the observed views by a single objective combining projection-domain reconstruction with regularization on the volumetric density and the K-Planes features:

$$\mathcal{L} = \mathcal{L}_1 + \lambda_{\text{d}}\mathcal{L}_{\text{dssim}} + \lambda_{\text{v}}\mathcal{L}_{\text{3DTV}} + \lambda_{\text{p}}\mathcal{L}_{\text{pTV}}. \quad (11)$$

The first two terms are the standard L_1 and D-SSIM losses between rendered and observed projections; $\mathcal{L}_{\text{3DTV}}$ is a total-variation prior on the volumetric density retrieved by the differentiable voxelizer of R2-Gaussian, and $\mathcal{L}_{\text{pTV}}$ is a planar total-variation prior on the K-Planes features used by ADM. The K-Planes features and the MLP parameters of ADM are jointly updated with the backbone Gaussian parameters under $\mathcal{L}$, so density modulation is learned online rather than fixed in advance.

4. Experiments

4.1. Experimental Settings

Dataset. Following NAF [GWS*24] and X-Gaussian [CLW*24], we adopt the public datasets of

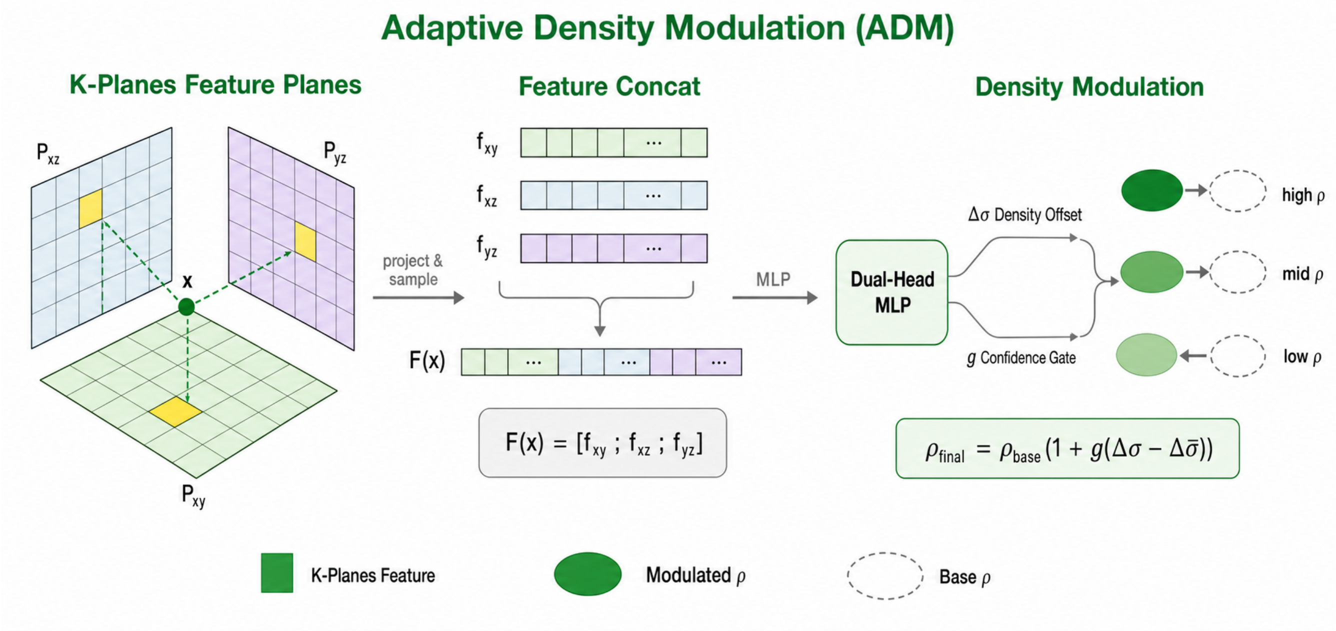

Figure 5: ***Adaptive Density Modulation (ADM).*** *For each Gaussian center, three orthogonal K-Planes feature planes are bilinearly sampled and concatenated; a dual-head MLP outputs a density offset and a confidence gate, which together apply a batch-centered relative correction* $\rho_{final} = \rho_{base}(1 + g(\Delta\sigma - \overline{\Delta\sigma}))$.

human organ CTs, *i.e.*, LIDC-IDRI and the open scientific visualization dataset, to evaluate our method. The test scenes include Chest, Foot, Head, Abdomen, and Pancreas. We adopt the open-source tomographic toolbox TIGRE to capture 100 projections with 3% noise in the range of $0 \sim 180°$. In the sparse-view NVS task, we retain only 2, 3, or 4 projections for training, and the remaining projections are used for testing. The CT volumes are used for testing in the sparse-view CT reconstruction task.

Baselines. We compare **RAttAGS** with six state-of-the-art methods from two categories. *X-ray/CT reconstruction methods:* X-Gaussian [CLW*24] [ECCV'24] introduces radiative Gaussian splatting for X-ray NVS; R2-Gaussian [ZLC*24] [NeurIPS'24] rectifies the integration bias in 3DGS and develops a differentiable voxelizer for tomographic reconstruction; and X-Field [WTW*25] [NeurIPS'25] employs physically informed ellipsoids with material-aware optimization. *Natural-scene sparse-view 3DGS methods:* CoR-GS [ZLY*24] [ECCV'24] introduces co-regularization to stabilize sparse-view training; DNGaussian [LZB*24] [CVPR'24] injects depth normalization for geometric guidance; and FSGS [ZFJW24] [ECCV'24] augments densification with proximity-guided unpooling. All methods are evaluated under the same sparse-view protocol on the same dataset.

Metrics. We adopt peak signal-to-noise ratio (PSNR) and structural similarity index measure (SSIM) to evaluate the performance. Both metrics are computed on $[0, 1]$-normalized projections, averaged over all held-out views of each scene and then over the five organs.

Implementation Details. Our **RAttAGS** is implemented in PyTorch and CUDA. The model is trained with the Adam optimizer for 3×10^4 iterations. SPS uses mixture coefficient $\alpha = 0.2$, power exponent $\gamma = 1.0$, and 50K initial Gaussians. GAP uses $K = 5$ nearest neighbors, proximity threshold $\tau = 0.015$, maximum pruning ratio $\beta_{\text{prune}} = 2\%$, and gradient threshold $\delta = 0.0002$. ADM uses K-Planes at resolution 64×32 and activates after 15K iterations. All experiments are conducted on a single NVIDIA RTX A6000 GPU.

4.2. Comparison with State-of-the-Art Methods

Discussion on NVS Quantitative Results. Tables 1 and 2 report the PSNR and SSIM comparisons between our **RAttAGS** and six state-of-the-art methods on the NVS task under the 2-view, 3-view, and 4-view sparse settings. The natural-scene sparse-view 3DGS methods (CoR-GS, DNGaussian, FSGS) struggle under the X-ray imaging regime, since their densification strategies and geometric priors are designed for visible-light reflection and scattering rather than X-ray penetration and attenuation. Among the X-ray-specific methods, X-Gaussian achieves reasonable performance but remains limited by the lack of integration bias correction. R2-Gaussian provides strong baseline results through its rectified rasterization and differentiable voxelizer. Our **RAttAGS** achieves the best average PSNR and SSIM across all three sparsity levels, ranking first in 12 out of 15 per-organ PSNR cells and 9 out of 15 SSIM cells. The margin over R2-Gaussian is most pronounced under the 2-view setting (+1.55 dB PSNR), indicating that attenuation-aligned density control is especially effective in the most severely under-determined regime where capacity misallocation is most damaging; the advantage persists at 3-view (+0.34 dB) and 4-view (+0.38 dB). X-Field shows competitive results in the human organ reconstruction setting, but its material-based optimization faces challenges under extremely sparse views where material boundaries are ambiguous.

Discussion on NVS Qualitative Results. Figure 6 depicts the qualitative comparisons of NVS on four representative organs (Chest, Head, Foot, and Pancreas) under the 2-view, 3-view, and 4-view settings. Under highly sparse view settings, all methods face challenges from limited input information. DNGaussian produces noticeable blurry artifacts due to its depth-normalization scheme being misaligned with the X-ray transmission model. R2-Gaussian reconstructs the overall anatomical structure better than the natural-scene baselines, but still exhibits boundary over-accumulation artifacts, particularly visible as streak patterns near high-contrast bone–soft-tissue interfaces. X-Field produces competitive results with reduced artifacts in organ interiors, though under the 2-view setting it introduces wave-pattern artifacts in the head scene. Our **RAttAGS** effectively mitigates these artifacts by reallocating Gaussian capacity from over-densified boundaries to interior attenuation-relevant structures. In particular, **RAttAGS** recovers smoother soft-tissue regions in the Chest, sharper bone details in the Foot, and cleaner background in the Head, while maintaining structural consistency across all three view budgets.

4.3. Ablation Study

Component Analysis. We conduct a break-down ablation experiment to study the effect of each proposed module. We adopt the R2-Gaussian backbone as the baseline model. The results are listed in Table 3. We can observe from the table: **(i)** When adding SPS for initialization, the performance yields improvement especially under the most under-determined 2-view regime, establishing that path-

Table 1: *PSNR comparison on the novel view synthesis task. We colorize the best, second-best, and third-best numbers. All methods are evaluated under the unified five-organ protocol.*

Method	Chest			Head			Abdomen			Foot			Pancreas			Average		
	2-view	3-view	4-view	2-view	3-view	4-view	2-view	3-view	4-view	2-view	3-view	4-view	2-view	3-view	4-view	2-view	3-view	4-view
Natural-scene sparse-view 3DGS methods																		
CoR-GS [ZLY*24] [ECCV'24]	14.96	18.55	20.29	16.24	20.36	23.40	18.50	22.45	23.03	20.17	25.53	26.83	16.15	23.25	23.96	17.20	22.03	23.50
DNGaussian [LZB*24] [CVPR'24]	18.84	20.55	20.99	15.72	18.04	16.24	14.30	16.02	15.76	19.33	23.51	24.72	19.06	22.43	25.47	17.45	20.11	20.64
FSGS [ZFJW24] [ECCV'24]	17.86	19.99	21.03	19.43	21.53	24.91	18.75	23.34	23.63	22.17	25.37	27.08	23.03	25.27	26.22	20.25	23.10	24.57
X-ray/CT reconstruction methods																		
X-Gaussian [CLW*24] [ECCV'24]	17.79	20.29	20.90	19.62	21.52	24.46	18.93	23.54	23.85	22.34	25.48	26.92	23.21	25.12	25.55	20.38	23.19	24.34
R2-Gaussian [ZLC*24] [NeurIPS'24]	21.05	26.22	25.47	23.54	26.60	28.66	24.85	29.26	30.56	19.39	28.48	30.04	17.83	28.61	30.90	21.33	27.83	29.13
X-Field [WTW*25] [NeurIPS'25]	19.57	23.94	24.45	22.49	25.06	26.78	22.20	24.87	27.19	20.45	26.03	27.25	18.12	24.94	26.36	20.57	24.97	26.41
RAttAGS (Ours)	21.22	26.63	26.03	24.26	26.65	29.68	25.26	29.55	30.90	20.82	28.64	29.95	22.99	29.37	30.95	22.91	28.17	29.50

Table 2: *SSIM comparison on the novel view synthesis task. We colorize the best, second-best, and third-best numbers. All methods are evaluated under the unified five-organ protocol.*

Method	Chest			Head			Abdomen			Foot			Pancreas			Average		
	2-view	3-view	4-view	2-view	3-view	4-view	2-view	3-view	4-view	2-view	3-view	4-view	2-view	3-view	4-view	2-view	3-view	4-view
Natural-scene sparse-view 3DGS methods																		
CoR-GS [ZLY*24] [ECCV'24]	0.5754	0.7077	0.7636	0.7993	0.8644	0.9088	0.8645	0.9089	0.9269	0.6913	0.8511	0.8767	0.8088	0.8974	0.9084	0.7479	0.8459	0.8769
DNGaussian [LZB*24] [CVPR'24]	0.6675	0.7500	0.7791	0.7296	0.8097	0.7924	0.7663	0.8258	0.8486	0.7801	0.8569	0.8666	0.8082	0.8848	0.8982	0.7503	0.8254	0.8370
FSGS [ZFJW24] [ECCV'24]	0.6451	0.7429	0.7729	0.8334	0.8732	0.9105	0.8580	0.9068	0.9276	0.6843	0.8500	0.8760	0.8564	0.9031	0.9148	0.7754	0.8552	0.8804
X-ray/CT reconstruction methods																		
X-Gaussian [CLW*24] [ECCV'24]	0.6488	0.7427	0.7689	0.8347	0.8697	0.9079	0.8595	0.9077	0.9257	0.6882	0.8481	0.8740	0.8502	0.9006	0.9124	0.7763	0.8538	0.8778
R2-Gaussian [ZLC*24] [NeurIPS'24]	0.7110	0.8367	0.8546	0.8718	0.9223	0.9550	0.9028	0.9363	0.9617	0.6722	0.8977	0.9130	0.8122	0.9199	0.9360	0.7940	0.9026	0.9241
X-Field [WTW*25] [NeurIPS'25]	0.6843	0.8090	0.8255	0.8413	0.8878	0.9089	0.7190	0.8632	0.9103	0.6522	0.8405	0.8556	0.6871	0.8474	0.9004	0.7168	0.8496	0.8801
RAttAGS (Ours)	0.7065	0.8390	0.8586	0.8694	0.9176	0.9518	0.9068	0.9374	0.9627	0.6910	0.8993	0.9149	0.8238	0.9247	0.9361	0.7995	0.9036	0.9248

anchored initialization from the coarse FDK attenuation support provides a meaningful head start for optimization. **(ii)** Adding GAP yields a consistent PSNR gain by reclaiming capacity from over-densified boundary regions, and contributes the highest SSIM among the individual modules, confirming that geometry-aware pruning effectively corrects the capacity misallocation inherited from visible-light densification. **(iii)** Adding ADM provides further refinement through position-dependent density correction, particularly benefiting scenes with heterogeneous attenuation profiles. The full **RAttAGS** framework, combining all three modules, achieves the best performance across all view counts.

Parameter Analysis. We further investigate the sensitivity of one representative hyperparameter from each module in Table 3. **(i)** The SPS mixture coefficient α balances the FDK-derived attenuation support against random initialization; $\alpha = 0.2$ offers sufficient path-anchored guidance without over-constraining the initial distribution. **(ii)** The GAP proximity threshold τ controls how aggressively crowded boundary Gaussians are pruned; a larger τ retains more crowded Gaussians and reduces the pruning effect, while a smaller τ prunes more aggressively and risks removing useful Gaussians near legitimate boundary structures. **(iii)** The ADM activation iteration determines when density modulation is introduced; among the tested schedules, earlier activation (Iter = 0) achieves the highest PSNR, and performance remains stable across the 10K–20K range, while activating too late (Iter = 25K) leaves insufficient iterations for the modulation to take effect. The

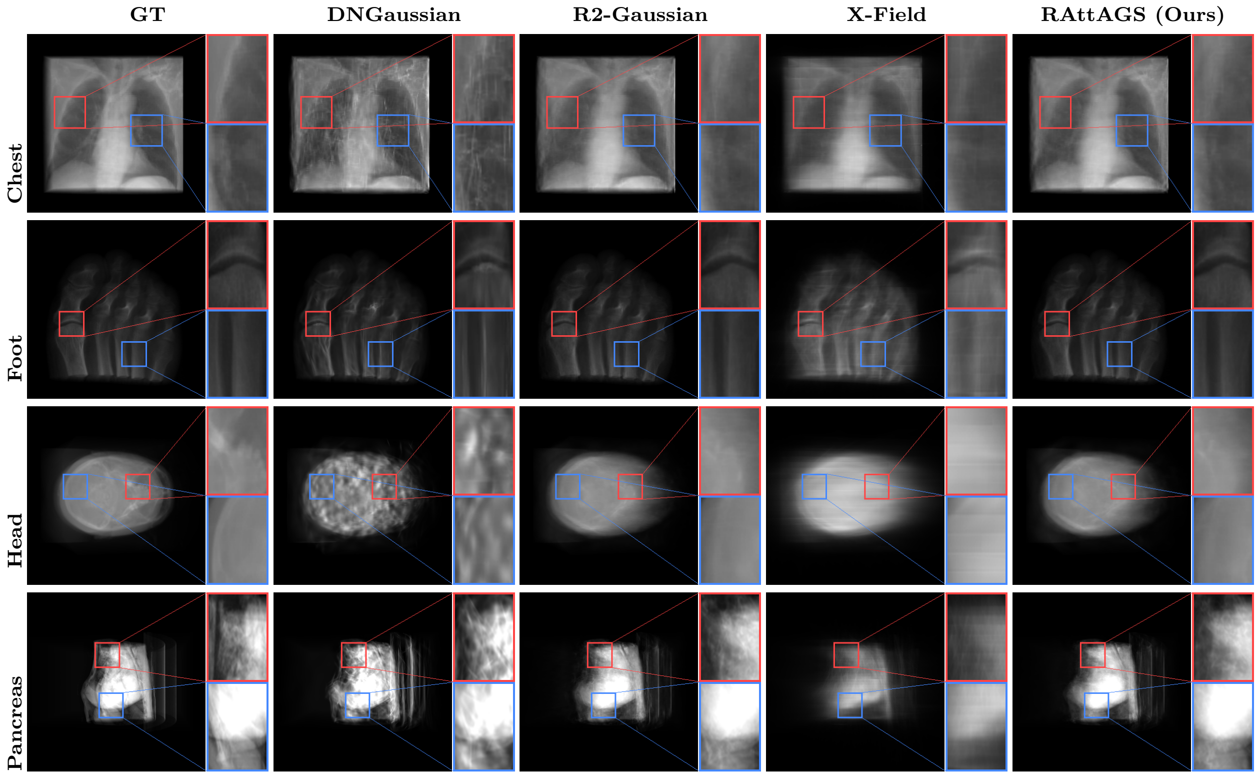

Figure 6: ***Qualitative comparison of NVS.*** *We present visual examples of rendered X-ray projections across four organs (Chest, Foot, Head, Pancreas). Columns: GT, DNGaussian, R2-Gaussian, X-Field, and* ***RAttAGS*** *(Ours). Our method produces clearer structural details, smoother textures in homogeneous regions, and fewer artifacts near anatomical boundaries. Red and blue boxes highlight zoomed-in regions of interest.*

default configuration ($\alpha = 0.2$, $\tau = 0.015$, ADM activation at 15K) achieves robust performance across the swept range, demonstrating that **RAttAGS** is not sensitive to individual hyperparameter choices.

Table 3: *Ablation results with our choices in* ***bold****. We colorize the best, second-best, and third-best numbers.*

Configuration	PSNR↑	SSIM↑
Component analysis (5-organ avg, 3-view)		
Baseline (R2-Gaussian)	27.83	0.9026
B + SPS	28.14	0.9032
B + GAP	28.10	0.9037
B + ADM	28.06	0.9041
Full RAttAGS	28.17	0.9036
SPS mixture coefficient α (5-organ avg, 3-view)		
$\alpha = 0.0$	28.09	0.9036
$\alpha = 0.1$	28.13	0.9034
$\boldsymbol{\alpha = 0.2}$	28.17	0.9036
$\alpha = 0.5$	28.11	0.9043
$\alpha = 1.0$	28.06	0.9044
GAP proximity threshold τ (5-organ avg, 3-view)		
$\tau = 0.005$	27.86	0.9030
$\tau = 0.010$	28.28	0.9041
$\boldsymbol{\tau = 0.015}$	28.17	0.9036
$\tau = 0.020$	28.22	0.9041
$\tau = 0.030$	28.15	0.9050
ADM activation iteration (5-organ avg, 3-view)		
Iter = 0	28.17	0.9036
Iter = 5K	28.09	0.9043
Iter = 10K	28.11	0.9042
Iter = 15K	28.13	0.9044
Iter = 20K	28.16	0.9045
Iter = 25K	28.04	0.9041

5. Conclusion

We have presented **RAttAGS**, a framework that identifies and corrects the capacity misallocation arising when adaptive density control designed for visible-light reflection and scattering is inherited by transmissive X-ray imaging. To anchor Gaussians along physically meaningful attenuation paths from the outset, SPS seeds initialization from a coarse volumetric support estimate; to drain the boundary-driven redundancy that gradient growth continuously produces, GAP reclaims locally crowded Gaussians during densification; and to resolve the heterogeneous density landscape that a single per-Gaussian gradient step cannot capture, ADM applies position-dependent modulation informed by spatial context. Together, these three attenuation-aligned stages redirect representation capacity from over-densified anatomical interfaces toward the interior structures that govern line-integral attenuation, yielding consistent improvements over state-of-the-art methods across diverse organs and sparse-view budgets. Because the framework augments rather than replaces the standard densification backbone, it is in principle applicable to other transmissive modalities—such as industrial CT, neutron imaging, or translucent-object reconstruction—where line-integral physics similarly demands capacity along the entire ray path rather than at surfaces alone.

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